HTML5, CSS Layout and Responsive Design

Web Development - Unit 1 - Structure and style

Semantic HTML

Semantic elements describe meaning, which helps accessibility, SEO and maintainability. Prefer header, nav, main, section, article, aside and footer over generic div elements.

```
<main>
  <article>
    <h1>Normalization</h1>
    <p>Remove redundancy using normal forms.</p>
  </article>
</main>
```

The CSS box model

Every element is a box: content, padding, border and margin. With box-sizing: border-box, the declared width includes padding and border, which makes layouts predictable.

Flexbox and Grid

- Flexbox is one-dimensional - great for navbars, toolbars and aligning items in a row or column.
- Grid is two-dimensional - great for page layouts and card galleries.

```
.cards {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(260px, 1fr));
  gap: 1rem;
}
```

Responsive design

- Design mobile-first, then add breakpoints with min-width media queries.
- Add <meta name="viewport" content="width=device-width, initial-scale=1">.
- Use relative units (rem, %, fr) and fluid images (max-width: 100%).

Accessibility checklist

- Every image has meaningful alt text; every form field has a label.
- Keep text contrast at or above 4.5:1 and make focus states visible.
- Everything usable by mouse must be usable by keyboard.